



White above 4 kHz: every spot reads the same. Two reasons. The foam absorbs short waves well (under 9 cm above 4 kHz), so little is reflected. And a band at 4 kHz is 900 Hz wide, so a reflection cancels and adds many times inside it and averages out; at 250 Hz the band is 58 Hz wide and holds one cancellation, so it shows. White means the arc and the microphones work; the problem is limited to the low bands.

Fig. 2. one fit over keys dp1-baseline-horizontal-prop7...prop18, PWM 1900, prop11-14 mirrored. rows aligned: mic row i sat at spot row i. 0° mic never moved, so its M and the 0° P are known only as a sum. third-octave centre, Hz

- Put a small loudspeaker on the stand instead of the propeller. Play short tone bursts at 250 and 800 Hz and record all microphones. The delay of the second arrival at each microphone, times 343 m/s, is the extra path length. Only one surface fits all eleven delays. With no airflow, anything left is the room.
- Write down the geometry in docs/mic_arc.md: arc radius, propeller height, which wall or corner is at the -90° end, distances from the -72° and $+72^\circ$ microphones to the nearest wall.
- Cover or move that surface, then run scripts/arc_error_map.py again. Foam only helps at 3 kHz. At 250 Hz the only fix is distance.

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4; prop18 with 811-1892 reads +3.1 = +4.7 - 4.2. Two mics, one room value.